free action : if $\exists\, p \in P$ $\quad p \triangleleft g = p \quad \Rightarrow \quad g = e$

$M = P/G = \{ [p] \mid p \in P \}$ $\qquad [p] = \{ p \triangleleft g \mid g \in G \}$ orbit of p

$\pi : p \in P \mapsto [p]$

<u>with other approach</u>

$\pi : P \to M$ fibre bundle, $F = G$, with G-atlas $\{(U_i, \phi_i)\}$

$\phi_{ij} = \phi_i \circ \phi_j^{-1} : (u, g) \in (U_i \cap U_j) \times G \mapsto (u, t_{ij}(u) \triangleright g) \in (U_i \cap U_j) \times G$

where $t_{ij} : U_i \cap U_j \to G$ smooth

define action of G on P as:

$(u, h) \in U \times G$

$(u, h) \triangleleft g = (u, hg)$

if $p \in \pi^{-1}(U_i)$ then $p \triangleleft g = \phi_i^{-1}(\phi_i(p) \triangleleft g)$

$\phi_i(p \triangleleft g) = \phi_i(p) \triangleleft g$ equivariant action

$\boxed{\text{action is transitive}}$ $\Rightarrow$ means if $p, q \in F_a = \pi^{-1}(\{a\})$ then there is $g \in G$ s.t. $q = p \triangleleft g$

① action is fibre-preserving : $p \in F_a \Rightarrow \phi_i(p) \in \{a\} \times G$

$\phi_i(p) \triangleleft g \in \{a\} \times G$

$p \triangleleft g = \phi_i^{-1}(\phi_i(p) \triangleleft g) \in F_a$

② $p, q \in F_a \Rightarrow \exists\, g_1, g_2 \in G$ s.t. $p = \phi_i^{-1}(a, g_1)$

$q = \phi_i^{-1}(a, g_2)$

action of G on itself is transitive: $\exists\, g$ s.t. $g_2 = g_1 g$ $\quad (g = g_1^{-1} g_2)$

$q = \phi_i^{-1}(a, g_2) = \phi_i^{-1}(a, g_1 g) = \phi_i^{-1}(a, g_1) \triangleleft g = p \triangleleft g$

difference between approaches: (A) (my approach) (B) Kobayashi:

(A) M is a fundamental object (spacetime)

$M \cong P/G$ (as a bundle)

(B) $M = P/G$

action of G on P is given

$$\Phi : P \times G \to P \qquad \text{group action}$$
$$(p, g) \mapsto p \triangleleft g$$

one-parameter subgroup: look at $p \triangleleft e^{tA}$ $\quad A \in \underline{\mathfrak{g}}$

define a vector field on P associated to A

define a vector at $u \in P$ as a linear map $X : \mathcal{C}^\infty(M, \mathbb{R}) \to \mathbb{R}$

that satisfies the product rule $\quad X(fg) = f(u)\, X(g) + g(u)\, X(f)$

$A^*_u \in T_u P$ define through its action as a derivation

$$A^*_u f = \frac{d}{dt}\left(f(u \triangleleft e^{tA}) \right)\Big|_{t=0}$$

$$A^* : P \to TP \qquad \text{vector field}$$
$$u \mapsto (u, A^*_u)$$

$$A^*_u (fg) = (A^*_u f)\, g(u) + (A^*_u g)\, f(u)$$

"fiber" vector field $M \to TM$? (section of TM)

$\sigma : \underset{\text{open}}{U \subseteq M} \to E$ ← some fibre bundle $\quad$ is a section if $\pi \circ \sigma = \mathrm{id}$

maps $u \in U$ to F_u

<u>example</u> action of $SO(2)$ on $\mathbb{R}^2$

$$R(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \qquad R(\theta) \triangleright (x, y) = \begin{pmatrix} \cos\theta\, x + \sin\theta\, y \\ -\sin\theta\, x + \cos\theta\, y \end{pmatrix}$$

only one basis element of $\underline{SO(2)}$: $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$R(\theta) = e^{\theta J}$

$$J^*_{(x,y)} f = \frac{d}{d\theta}\Big|_{\theta=0} f(\cos\theta\, x + \sin\theta\, y,\ -\sin\theta\, x + \cos\theta\, y)$$

$$= \frac{\partial f}{\partial x}(x,y)\cdot y - \frac{\partial f}{\partial y}(x,y)\cdot x$$

$J^* = (y,-x)\cdot\nabla$ dir. derivative in dir $(y,-x)$

---

if X vector field, it defines a 1st order ODE

$\gamma : t \in I \underset{open}{\subseteq} \mathbb{R} \to \gamma(t) \in M$

$\dot\gamma(t)$ is a vector in $T_{\gamma(t)}M$ defined by action on functions

$\dot\gamma(t) f = (f\circ\gamma)'(t)$ ← dir. derivative

ODE $\begin{cases} \dot\gamma(t) = X_{\gamma(t)} \\ \gamma(t_0) = P_0 \in M \end{cases}$ ← solution exists + unique

γ is called integral curve of X

on $\mathbb{R}^n$ ODE

$F : \mathbb{R}^n \to \mathbb{R}^n$

$\begin{cases} \frac{d}{dt}\gamma(t) = F(\gamma(t)) \\ \gamma(t_0) = P_0 \end{cases}$

$\Updownarrow$

$\begin{cases} \gamma'(t)\cdot\nabla = F(\gamma(t))\cdot\nabla \\ \gamma(t_0) = P_0 \end{cases}$

---

since sol. is unique $\begin{cases} \dot\gamma(t) = A^*_{\gamma(t)} \\ \gamma(0) = u \end{cases}$

$\frac{d\gamma^1}{dt}\cdot\frac{\partial}{\partial x} + \frac{d\gamma^2}{dt}\cdot\frac{\partial}{\partial y}$

$\shortparallel$

$F^1(\gamma(t))\frac{\partial}{\partial x} + F^2(\gamma(t))\frac{\partial}{\partial y}$

$\gamma(t) = u \triangleleft e^{tA}$ → $\dot\gamma(t) f = (f\circ\gamma)'(t)$

$= \frac{d}{dt} f(u \triangleleft e^{tA}) = A^*_{\gamma(t)}$

A^*_u is tangent to the fibre. What does it mean?

if γ is the integral curve passing through u

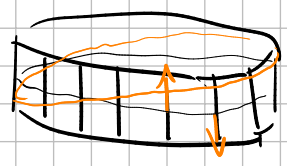
the γ is contained in the fibre

it is because $\gamma(t) = u \underbrace{\triangleleft e^{tA}}_{\text{preserves fibre}}$ → if $u \in F_a$

then $\gamma(t) \in F_a \ \forall t$

A^* is a <u>vertical</u> vector field

↳ tangent to the fibre at each point



$\gamma(t) = u \triangleleft e^{tA}$

$$\dot{\gamma}(t) f = (f \circ \gamma)'(t) = \frac{d}{dt} f(u \triangleleft e^{tA}) \overset{?}{=} A^*_{\gamma(t)} f$$

$$A^*_{\gamma(t)} f = \frac{d}{ds}\Big|_{s=0} f(\gamma(t) \triangleleft e^{sA})$$

$$= \frac{d}{ds}\Big|_{s=0} f(u \triangleleft e^{tA} \triangleleft e^{sA})$$

$$= \frac{d}{ds}\Big|_{s=0} f(u \triangleleft e^{(t+s)A})$$

$$= \frac{d}{dt} f(u \triangleleft e^{tA})$$